Imputing Multi-Agent Trajectories from Event and Snapshot Data in Soccer

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Abstract

Recent advances in wearable sensors and computer vision technologies have enabled the collection of tracking data in team sports, which has become a core resource for fine-grained analysis. However, the availability of tracking data remains constrained by high acquisition costs and technical limitations. Compared to tracking data, event data recording on-ball actions and snapshot data providing partial player positions are more widely accessible. To this end, this study proposes a novel approach to predict the positions of all players at each event timestamp in soccer matches, leveraging the limited information available from event and snapshot data. We propose an event-based imputation model that integrates spatial and temporal attention to capture the spatiotemporal multi-agent structure. In experiments, we evaluate our model on 13 soccer matches, achieving average position errors of 5.84 m. To assess the practical utility of our approach, we apply it to a downstream task called Pitch Control, which requires full tracking data. These results highlight the potential of event-based position imputation to expand access to fine-grained analysis in data-constrained settings.

CCS Concepts

• **Computing methodologies** → **Neural networks**.

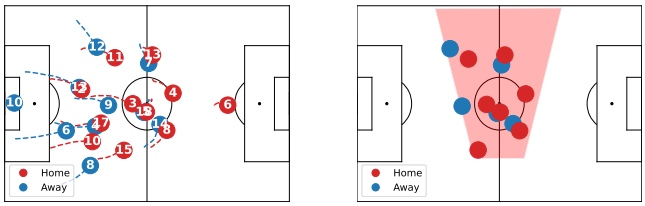
Keywords

Time-Series Data Imputation, Sports Data Analytics, Multi-Agent Systems, Spatio-Temporal Modeling, Event-Based Data

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(a) Full tracking data **(b) Snapshot data**

Figure 1: Comparison between full tracking data and snapshot data. (a) Full tracking data continuously records the positions of all players throughout the match. (b) Snapshot data intermittently captures the positions and role flags (i.e., teammate, actor, and keeper flags) of a subset of players visible within the camera's field of view at event timestamps.

1 Introduction

The problem of trajectory prediction in multi-agent systems has attracted significant attention as a crucial task across various domains, including autonomous driving [14, 23], pedestrian trajectory forecasting [1, 10, 19], and general multi-agent settings [3, 30]. In particular, there has been growing interest in applying trajectory prediction methods to team sports such as soccer to forecast the future movements of players for tactical analysis [8, 9, 17].

In soccer matches, advances in wearable sensors and computer vision technologies have enabled the collection of tracking data. However, collecting full tracking data that includes the 2D positions of all players throughout the entire match remains challenging due to high costs and technical limitations. For these reasons, the data for research purposes is more often available in the form of incomplete snapshot representations, which are extracted from broadcast camera footage. Figure 1 illustrates the difference between full tracking data and snapshot data. While full tracking data contains complete spatiotemporal information of all players throughout the entire match, snapshot data only includes the positions and role flags (e.g., teammate flag) of a subset of players at specific moments, making it unsuitable for direct application to tactical analyses.

Compared to full tracking data, soccer event data—which records key ball-related actions such as shots and passes—is known to be easier to collect and more widely available across matches and

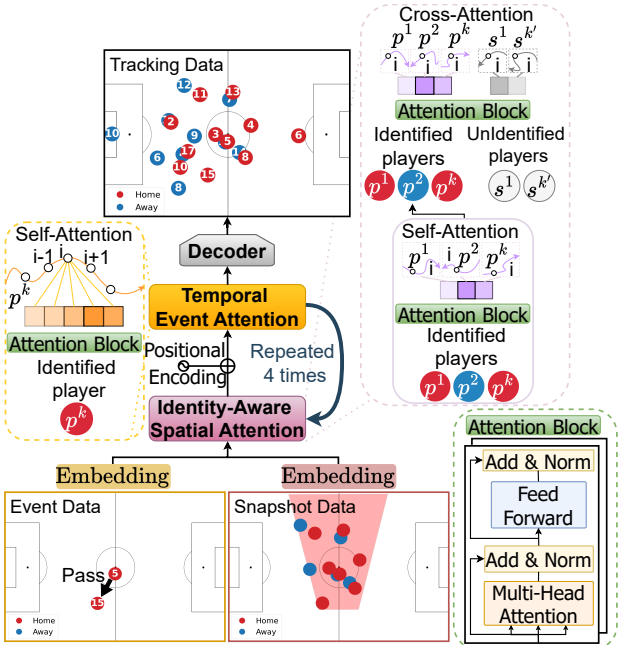


Figure 2: Overview of the proposed imputation model.

leagues since it can be collected fully remotely [16]. Taking advantage of this cost-efficient nature of event data, a pioneering study [8] proposed an imputation model called AgentImputer, which reconstructs all missing player positions using only event data. Specifically, AgentImputer is composed of LSTM [11] and GNN [20] modules, which are designed to learn temporal dynamics and inter-agent interactions, respectively. Although AgentImputer demonstrated the feasibility of imputing missing players' trajectories from limited data, the resulting position errors are still insufficient for precise tactical analysis. A key reason for this limitation lies in the sparsity of the input, as the event data provides only the positions of the players involved in each action (see Figure 2).

To address these limitations, we propose a model that integrates event data with snapshot data, incorporating a *spatial attention* mechanism to capture spatial interactions among players and a *temporal attention* mechanism to model the temporal dynamics of each individual player. Furthermore, we show that the predicted positions are effective for downstream tasks such as Pitch Control [22], suggesting that our approach is a practical alternative when high-resolution tracking data is unavailable.

2 Related Work

The multi-agent trajectory imputation task is to reconstruct unobserved positions of agents based on both past and future observations. This approach is particularly valuable in scenarios where observation data is partially missing.

Early research in trajectory imputation primarily employed RNN-based models to impute missing frames [5, 29]. A subsequent approach employs a GNN to model spatial dependencies and uncertainty [15, 26]. Due to the permutation-invariant nature of GNN, it has been widely applied to model spatiotemporal multi-agent settings such as soccer [2, 18, 28] and other team sports [21, 31]. More recent work incorporates attention mechanisms [27] to capture both temporal dependencies and inter-agent interactions [6, 7, 13].

However, these methods can only deal with data that is uniformly and continuously collected at each time step for every agent, limiting their applicability to inputs that are irregularly sampled and provide information restricted to specific agents, such as soccer event data. To this end, recent studies on soccer analysis have explored methods for imputing missing player positions by leveraging limited information [8, 12, 25].

3 Methodology

Our goal is to capture diverse information relevant to player behavior by constructing a structured input for the model. The model input consists of features extracted from event and snapshot data, categorized into four types as follows:

- **Spatial features**: Include not only the observed positions at each event timestamp, but also the previous and subsequent coordinates.
- **Contextual features**: Include match context (i.e., average ball possession and goal difference) and spatial arrangements from snapshot data, which consists of anonymized coordinates and role flags (i.e., teammate, actor, and keeper flags).
- **Temporal features**: Include temporal cues such as the time elapsed since a player was last observed and the time remaining until their next observation.
- **Categorical features**: Include categorical information about the action type (e.g., pass) and each player's role (e.g., striker).

3.1 Model Architecture

Figure 2 illustrates the overall architecture of the proposed imputation model, which comprises four components: a Multi-Modal Input Encoder, Identity-Aware Spatial Attention, Temporal Event Attention, and a Decoder. Each of these components will be described in the following subsections.

Multi-Modal Input Encoder. For the i-th event, let $P_i = \{p_i^1, \ldots, p_i^{K_i}\}$ denote the set of identified players, where K_i is the total number of players. Similarly, let $S_i = \{s_i^1, \ldots, s_i^{K_i'}\}$ denote the set of unidentified players within the camera, with $K_i' \leq K_i$. Each player $p^k \in P_i$ and $s^{k'} \in S_i$ is associated with a feature vector $x_i^{p^k} \in \mathbb{R}^{d_x}$ and $v_i^{s^{k'}} \in \mathbb{R}^{d_v}$, respectively (where d_x and d_v denote the dimensions of the feature vectors). These feature vectors are projected into embeddings using separate learnable functions $\phi_P : \mathbb{R}^{d_x} \to \mathbb{R}^{d'}$ and $\phi_S : \mathbb{R}^{d_v} \to \mathbb{R}^{d'}$, where d' is the embedding dimension:

$$z_i^I = (z_i^{p^1}, \ldots, z_i^{p^K}) = \phi_P(x_i^{p^1}, \ldots, x_i^{p^K}) \quad (1)$$

$$z_i^U = (z_i^{s^1}, \ldots, z_i^{s^{K'}}) = \phi_S(v_i^{s^1}, \ldots, v_i^{s^{K'}}) \quad (2)$$

To capture temporal context, we leverage a total of $2N + 1$ event frames centered at the target event e_i, where N denotes the number of frames preceding and succeeding event e_i. In summary, the embeddings in (1) and (2) produce tensors of shapes $(2N+1)\times K\times d'$ and $(2N + 1) \times K' \times d'$, respectively.

Identity-Aware Spatial Attention. We introduce a novel attention mechanism, termed *Identity-Aware Spatial Attention*. This module leverages the attention mechanism between two sets of spatial coordinates: 1) one with explicit player identities obtained from event data, and 2) the other from snapshot data without identity annotations. To capture interactions among identified players, this mechanism applies self-attention over the identified player embeddings (3). Subsequently, to solve the ambiguity of unidentified player positions in snapshot data, this mechanism applies attention from identity-known positions to identity-unknown positions (4). To further refine the alignment, identity-unknown positions also attend to identity-known positions (5). This enables the model to better align the snapshot coordinates with the known event-based positions, thereby improving the accuracy of position predictions.

$$H_i^{\mathrm{I}} = \mathrm{SelfAttention}(Q = K = V = z_i^I) \tag{3}$$

$$H_i^{\mathrm{U}\to\mathrm{I}} = \mathrm{CrossAttention}(Q = H_i^{\mathrm{I}}, K = V = z_i^U) \tag{4}$$

$$H_i^{\mathrm{I}\to\mathrm{U}} = \mathrm{CrossAttention}(Q = z_i^U, K = V = H_i^{\mathrm{I}}) \tag{5}$$

$$\tilde{H}_i = \mathrm{CrossAttention}(Q = H_i^{\mathrm{U}\to\mathrm{I}}, K = V = H_i^{\mathrm{I}\to\mathrm{U}}) \tag{6}$$

Temporal Event Attention. To capture the sequential movement patterns of identified players across event frames, we introduce a temporal attention mechanism, termed *Temporal Event Attention*. This module applies self-attention over the spatial encodings $\tilde{H}_i$ of each identified player independently across event frames.

$$\hat{H}_i = \mathrm{SelfAttention}(Q = K = V = \tilde{H}_i) \tag{7}$$

Position Decoder. To incorporate spatial context and facilitate accurate alignment with snapshot positions, we apply the Identity-Aware Spatial Attention and Temporal Event Attention modules iteratively (four times in our experiments). Finally, we use two fully connected layers to predict each player's position based on the refined encodings $\tilde{H}_i$.

$$\hat{y}_i = \mathrm{FC}(\tilde{H}_i), \tag{8}$$

3.2 Matching Algorithm

We apply a position correction strategy that leverages snapshot data to improve the alignment between predicted and ground-truth player positions. Specifically, we use the Hungarian algorithm to match the predicted positions ($\hat{y}^{p^k}$) with the anonymized positions ($y^{s^{k'}}$) for each team. We construct a cost matrix ($C \in \mathbb{R}^{K\times K'}$) where each element is the Euclidean distance between a predicted and anonymized position: $C_{k,k'} = \|\hat{y}^{p^k} - y^{s^{k'}}\|_2$.

By minimizing this cost matrix, the algorithm identifies the optimal matching. The matched predictions (K' of them) are then replaced with their corresponding anonymized positions, while any unmatched predictions ($K - K'$) remain unchanged.

4 Experiments

4.1 Datasets

We train and evaluate our model using event data and tracking data provided by the Deutsche Fußball Liga (DFL), including 7 publicly available matches [4] and 6 additional ones provided separately that are not publicly accessible. The data consists of event annotations and tracking data sampled at 25 Hz, collected by TRACAB (formerly Chyronhego) and Sportec Solutions following the official match data catalogue defined by the DFL. The snapshot data used as input to our model was generated from the tracking data by simulating a camera projection from an arbitrary location and extracting only the coordinates and role flags (i.e., teammate, actor, and keeper flags) of players visible within the camera's field of view. This generation process is designed to mimic the format of publicly available snapshot datasets, such as the StatsBomb360 data [24], ensuring the practical relevance of our approach. We train and evaluate our model on a dataset of 13 matches, comprising a total of 19,272 event records. We adopt a 13-fold cross-validation, using 10 matches for training, 2 for validation, and 1 for testing in each fold, ensuring that all matches are tested.

Table 1: Performance comparison of the baselines and the proposed model. The position error (PE) column represents the mean Euclidean distance between predicted and ground-truth positions. $\mathbf{PE_{match}}$ represents the mean Euclidean distance after applying the matching algorithm described in Section 3.2, while the accuracy (Acc) column represents the proportion of correctly matched players.

Model	Window	PE (m)	$\mathrm{PE_{match}}$ (m)	Acc (%)
Heatmap-based	Half	19.96 ± 1.00	13.99 ± 0.49	39.15 ± 1.89
XGBoost	5	9.92 ± 0.59	8.67 ± 0.64	47.73 ± 3.26
	11	9.84 ± 0.59	8.58 ± 0.65	47.97 ± 3.34
	21	9.87 ± 0.61	8.60 ± 0.65	47.94 ± 3.23
AgentImputer	5	8.66 ± 0.48	7.44 ± 0.50	54.79 ± 3.41
	11	8.74 ± 0.54	7.41 ± 0.62	53.65 ± 3.88
	21	9.65 ± 0.72	8.13 ± 0.71	49.05 ± 2.62
Ours (Event only)	5	7.76 ± 0.24	6.55 ± 0.29	61.85 ± 1.73
	11	7.59 ± 0.22	6.37 ± 0.28	62.59 ± 1.78
	21	7.61 ± 0.23	6.37 ± 0.28	62.55 ± 2.14
Ours (Event+Snapshot)	5	7.09 ± 0.26	6.01 ± 0.29	64.78 ± 2.27
	11	6.93 ± 0.26	5.87 ± 0.28	**65.52 ± 2.07**
	21	**6.91 ± 0.20**	**5.84 ± 0.24**	65.49 ± 2.32

4.2 Baselines

To evaluate the performance of the proposed method, we compare our model with several baseline models, including the AgentImputer [8], which is the state-of-the-art model in the proposed task.

- **Gaussian heatmap**: We represent the players' activity zones during a match using two-dimensional probability distributions (called heatmaps) generated under the Gaussian assumption.
- **XGBoost**: This model uses hand-crafted features derived from the event data as input to the XGBoost model.
- **AgentImputer**: This is the model proposed in a prior study [8], which encodes the temporal trajectories of individual players using LSTMs and learns player interactions using a GNN.

4.3 Implementation Details

The Heatmap-based method used a Gaussian distribution with a standard deviation of 15 for both the x and y axes, with a half-match window. For our proposed model, we set the embedding size to 128. The encoder was configured with a hidden dimension

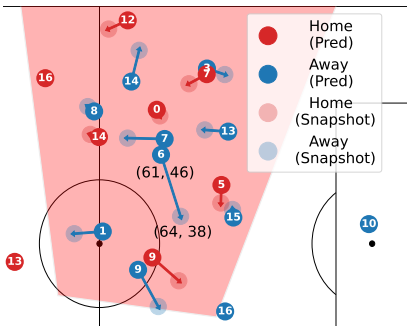


Figure 3: Examples of player matching results.

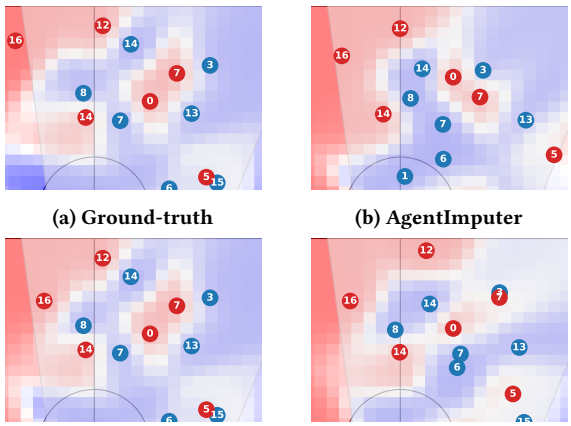


Figure 4: Pitch control computed based on player positions. Each region indicates a control probability ranging from 0 (blue) to 1 (red).

of 128, a multi-head attention with 16 heads, a single layer, and a dropout rate of 0.2. The decoder is a feed-forward network consisting of two linear layers with a GELU activation function, mapping the 128-dimensional hidden representations to two-dimensional coordinates.

4.4 Results and Analysis

Experimental results. Table 1 presents the performance comparison between the proposed model and the baseline methods. The heatmap-based and XGBoost-based methods utilize only individual player information and therefore fail to capture interactions between players, resulting in relatively poor performance. AgentImputer [8], which incorporates inter-player interactions using a GNN, achieves better performance in terms of position error. However, our reproduced implementation of the model yields lower performance than reported in [8], which can be attributed to the reduced size of the dataset (34 → 13 matches).

Effect of Temporal Event Attention. We show that our model exhibits improved performance as the temporal window size increases (e.g., 7.09 → 6.91 for window sizes 5 and 21, respectively). In contrast, all baseline models show negligible improvement in performance with increasing window size. These findings suggest that this mechanism effectively captures long-range match contexts.

Effect of Identity-Aware Spatial Attention. We evaluate the contribution of snapshot data by comparing the performance with and without its use. The results clearly demonstrate that incorporating snapshot data leads to substantial improvements in prediction accuracy (e.g., 62.59 → 65.52).

Effect of Matching Algorithm. We evaluate the contribution of the matching algorithm by comparing the performance with and without the algorithm described in Section 3.2. The results show that this algorithm consistently reduces the overall position error. Despite using a smaller dataset, our model achieved a PE_{match} of 6.37 with event-only data and 5.84 with the addition of snapshot data, both surpassing the best performance from a prior study (6.88).

4.5 Case Study

Figure 3 shows the matching results after applying the matching algorithm. For the 15 players (6 from the red team and 9 from the blue team) captured within the camera, the predicted positions were optimally matched with the anonymized positions. For instance, the predicted position $(61, 46)$ of blue team player #6 was found to have the highest matching confidence with the anonymized position $(64, 38)$, leading to a correction from $(61, 46)$ to $(64, 38)$.

4.6 Application

To demonstrate our approach's practical utility, we apply it to the downstream task of Pitch Control [22], a spatial analysis framework for estimating team dominance over different regions of the field. As shown in Figure 4, the pitch control maps generated by our model (4c and 4d) resemble the ground-truth maps more closely than AgentImputer's map (4b). Quantitatively, the average difference between our map and the ground-truth map is 0.073, compared to 0.106 for AgentImputer. After applying the matching algorithm, the error is further reduced to 0.026, demonstrating a substantial gain. These results highlight the effectiveness of our approach in enabling tactical analysis from limited data sources.

5 Conclusions

We have proposed a novel model that imputes missing player trajectories in soccer matches by leveraging widely accessible event and snapshot data, enabling fine-grained analysis of match dynamics without high-cost tracking data. Our approach opens up new possibilities for a range of downstream tasks, as demonstrated by our application to Pitch Control analysis. A key limitation is that the model does not incorporate player velocity, which restricts the scope for more diverse analyses. Addressing this limitation remains an important direction for future research.

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GenAI Usage Disclosure

Generative AI tools were utilized in limited stages of this work. Specifically, ChatGPT was employed to assist in drafting and refining parts of the manuscript text, and GitHub Copilot was used during the early development of code. All outputs generated by these tools were critically reviewed and validated by the authors. No generative AI tools were used for data collection, analysis, or interpretation.

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